

2026 VLSI Design and Implementation

LAB3 Clock Jitter

Objective

In this assignment, students will learn how to generate a clock signal with jitter and analyze its impact on the timing and performance of the overall design.

Due Date: 2026/5/15 12:00 p.m.

Lab3 Environment

This assignment will reuse the environment and the design from Lab 1.

You do not need to modify the RTL code of the design. You only need to modify the `PATTERN.v` provided by the TA and replace the original `PATTERN.v` file.

Task Description

You are required to complete the following tasks.

- **Clock Jitter Generation [40%]**

Modify the `PATTERN.v` to generate a clock signal that includes jitter. You must implement two different types of jitter distributions and explain your method in detail.

1. **Uniform Jitter:**

The total clock period **varies evenly and randomly** within the strict *range of `[CYCLE_TIME * 0.8, CYCLE_TIME * 1.2]`.

2. **Gaussian Jitter:**

The total clock period varies following a **bell-shaped normal distribution** centered at the ideal `CYCLE_TIME`.

***Jitter Limit:** The maximum variation (jitter) applied to the clock period must be bounded to **+20% of the ideal cycle time (`CYCLE_TIME`)**. The precision of the jitter must be at least 0.001. **You also need to explain how you met the jitter limit in the report.**

- **Limit Cycle Time Search [15%]**

Run 03 Gate-Level Simulation to **find the absolute minimum cycle time** (*Limit CT) where the design can still function correctly without any functional errors or timing violations (i.e. pass the pattern).

You need to find the Limit CT for three distinct scenarios:

1. Ideal Clock (Without Jitter)

2. Clock with Uniform Jitter
3. Clock with Gaussian Jitter

***Cycle Time Limit:** The precision of the clock period is 0.1, for example, 5.1 is allowed, 5.05 is not allowed.

You need to explain how you found the limit cycle time, and attach a screenshot of the passing 03 gate-level simulation in the report.

- **Clock with Jitter Visualization [20%]**

Record the generated clock periods during your simulation and use a tool of your choice to plot the **distribution histograms** of your generated Uniform and Gaussian jitters to verify their statistical characteristics.

(Hint: You can use some built-in Verilog function like “\$fdisplay” to write into a log file.)

You also need to explain how you collect the data.

- **Waveform Observation [10%]**

In this section, you must dump the waveform from your **Gate-Level Simulation with Gaussian Jitter** and capture the following two specific scenarios.

1. Jitter Verification (First 10 Cycles)

Screenshot: Capture a clear waveform screenshot showing the first 10 consecutive clock cycles of your simulation (with Gaussian Jitter).

Data Recording: In your report, explicitly list the exact duration (cycle time) of each of these first 10 cycles.

2. Failure Point Analysis

Simulation Setup: Set the baseline **CYCLE_TIME** to the Limit CT of the Ideal Clock (w/o jitter) that you discovered in the previous task. Since this baseline CT has zero timing margin, adding Gaussian jitter will inevitably cause timing violations.

Screenshot: Run the Gate-Level simulation **with Gaussian Jitter** and find the exact moment the design fails (e.g., the output mismatches the golden model, or output signals transition to a red **X** state due to setup/hold violations). Capture a screenshot of this failure point.

Data Recording: You must record and clearly state the exact clock cycle duration at the moment of failure. Briefly explain why this specific cycle length triggered the violation.

- Q&A [15%]

Answer the following questions in your report:

1. What are the physical causes of clock jitter in real IC chips?
2. What negative impacts does clock jitter bring to a digital system?
3. What are some common techniques or circuit designs used to reduce or solve clock jitter issues?

Grading Policy

1. Report: 100%

2. **Due Date: 2026/5/15 12:00 p.m.**

Please submit the following zip files to the E3 system before 12:00 at noon on May 15

- vlsixx.zip (ex. vlsi00.zip)
 - vlsixx (folder)
 - vlsixx_report.pdf
 - MATRIX_MULT.v
(You may submit the same file as Lab 1 if you do not modify it)
 - PATTERN.v

3. If uploaded files **violate the naming rule**, you will get **5 deduct points**.

Notes

- Report

You may follow the **Lab3_report_example** file to finish your report.
Both Chinese and English are acceptable.

- Late Submission

If the submission is made within one week after the due date, the score will be calculated as 80% of the original.

- How to dump waveform in 03 Gate-Level Simulation

- In **00_TESTBED/TESTBED.v**, uncomment the following code.
(Line 9~11 in picture)

```

1  initial begin
2    `ifdef RTL
3      $fsdbDumpfile("MATRIX_MULT.fsdb");
4      $fsdbDumpvars(0,"+mda");
5      $fsdbDumpvars();
6    `endif
7    `ifdef GATE
8      $sdf_annotate("MATRIX_MULT_SYN.sdf", u_MATRIX_MULT);
9      // $fsdbDumpfile("MATRIX_MULT_SYN.fsdb");
10     // $fsdbDumpvars(0,"+mda");
11     // $fsdbDumpvars();
12   `endif
13 end

```

- Copy 01_RTL/05_nWave file to 03_GATE/.
- Run 03_GATE/05_nWave after Gate-Level Simulation.